INCH-POUND
MIL-DTL-24215/9C
7 June 2002
SUPERSEDING
MIL-A-24215/9B
24 July 1980

DETAIL SPECIFICATION SHEET

# ATTENUATORS, VARIABLE (COAXIAL), CLASS 5 (SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the attenuators described herein shall consist of this specification and MIL-DTL-24215.

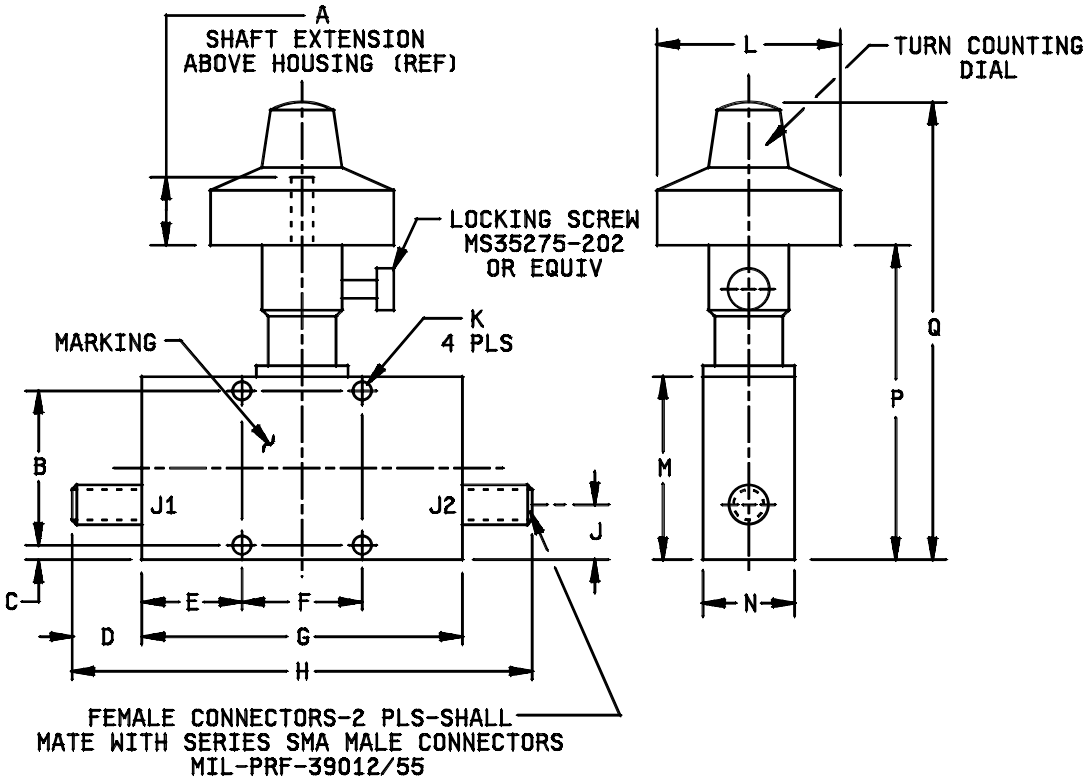



FIGURE 1. Dimensions and configuration for part number M24215/9-001.


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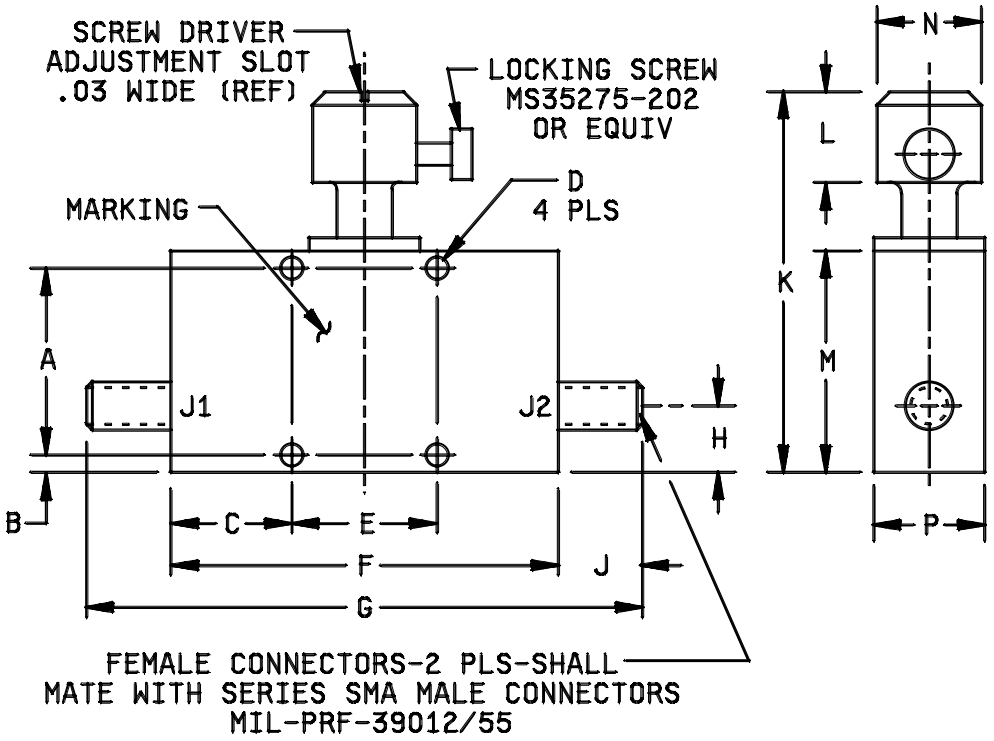
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | .58 | .54 | 14.7 | 13.7 |
| B | .849 | .839 | 21.56 | 21.31 |
| C | .11 | .06 | 2.8 | 1.5 |
| D | .40 | .36 | 10.2 | 9.1 |
| E | .58 | .52 | 14.7 | 13.2 |
| F | .661 | .651 | 16.79 | 16.54 |
| G | 1.78 | 1.72 | 45.2 | 43.7 |
| H | 2.54 | 2.48 | 64.5 | 63.0 |
| J | .33 | .27 | 8.4 | 6.9 |
| K | .107 dia | .101 dia | 2.72 dia | 2.57 dia |
| L | 1.03 dia | .97 dia | 26.2 dia | 24.6 dia |
| M | 1.03 | .97 | 26.2 | 24.6 |
| N | .53 | .47 | 13.5 | 11.9 |
| P | 1.75 | 1.69 | 44.4 | 42.9 |
| Q | 2.53 | 2.47 | 64.3 | 62.7 |

FIGURE 1. Dimensions and configuration for part number M24215/9-001 - Continued.



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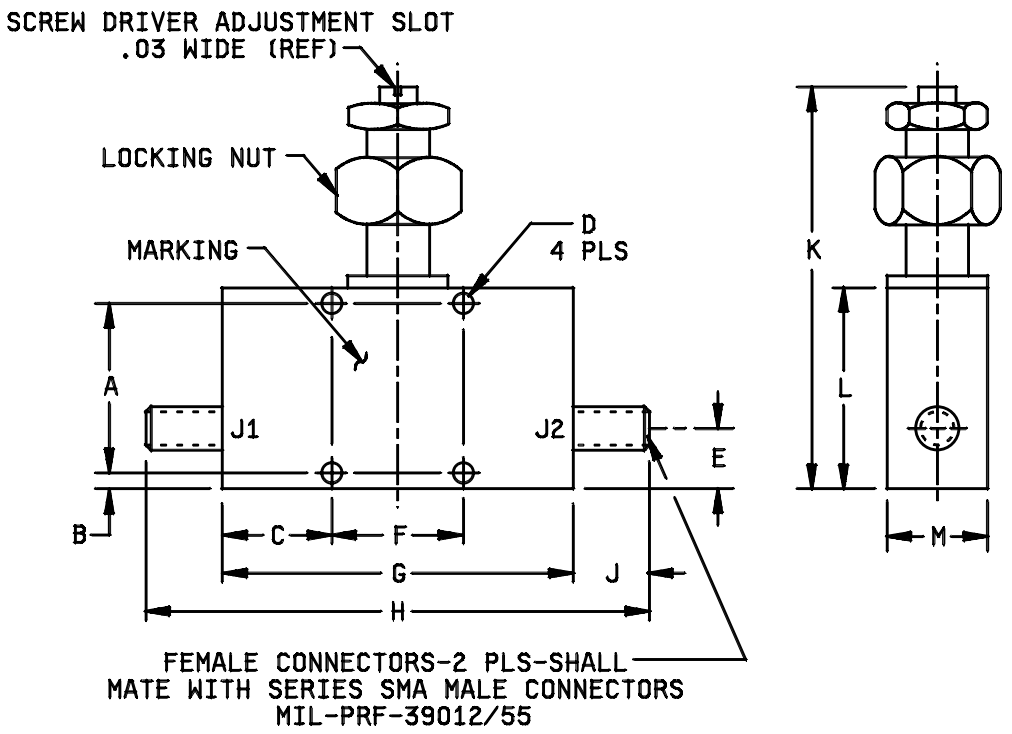

| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | .849 | .839 | 21.56 | 21.31 |
| B | .11 | .06 | 2.8 | 1.5 |
| C | .58 | .52 | 14.7 | 13.2 |
| D | .107 dia | .101 dia | 2.72 dia | 2.57 dia |
| E | .661 | .651 | 16.79 | 16.54 |
| F | 1.78 | 1.72 | 45.2 | 43.7 |
| G | 2.54 | 2.48 | 64.5 | 63.0 |
| H | .303 | .297 | 7.70 | 7.54 |
| J | .41 | .35 | 10.4 | 8.9 |
| K | 1.75 | 1.69 | 44.4 | 42.9 |
| L | .46 | .36 | 11.7 | 9.1 |
| M | 1.03 | .97 | 26.2 | 24.6 |
| N | .52 | .42 | 13.2 | 10.7 |
| P | .53 | .47 | 13.5 | 11.9 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Dimensions and configuration for part number M24215/9-002.



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| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | .849 | .839 | 21.56 | 21.31 |
| B | .11 | .06 | 2.8 | 1.5 |
| C | .58 | .52 | 14.7 | 13.2 |
| D | .107 dia | .101 dia | 2.72 dia | 2.57 dia |
| E | .303 | .297 | 7.70 | 7.54 |
| F | .661 | .651 | 16.79 | 16.54 |
| G | 1.78 | 1.72 | 45.2 | 43.7 |
| H | 2.54 | 2.48 | 64.5 | 63.0 |
| J | .40 | .36 | 10.2 | 9.1 |
| K | 2.03 | 1.97 | 51.6 | 50.0 |
| L | 1.03 | .97 | 26.2 | 24.6 |
| M | .53 | .47 | 13.5 | 11.9 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Dimensions and configuration for part number M24215/9-003.



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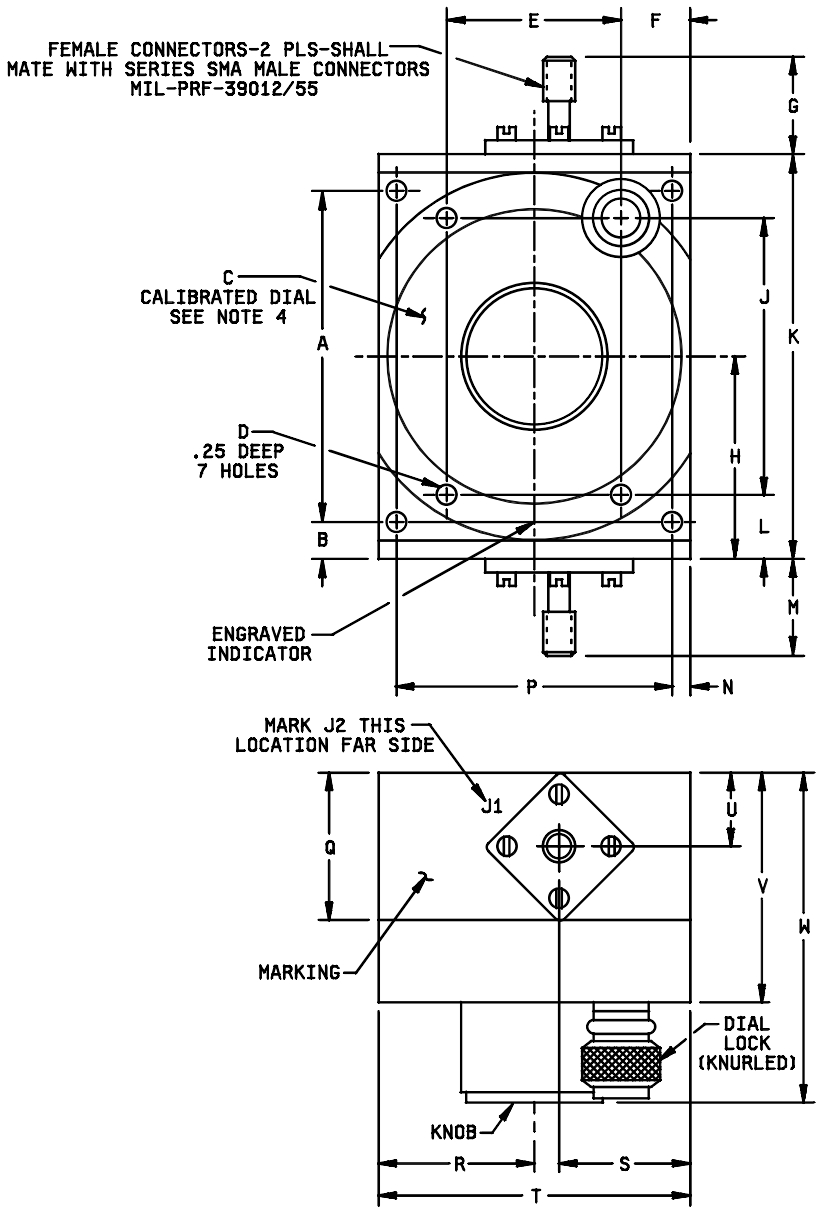



FIGURE 4. Dimensions and configuration for part number M24215/9-004.



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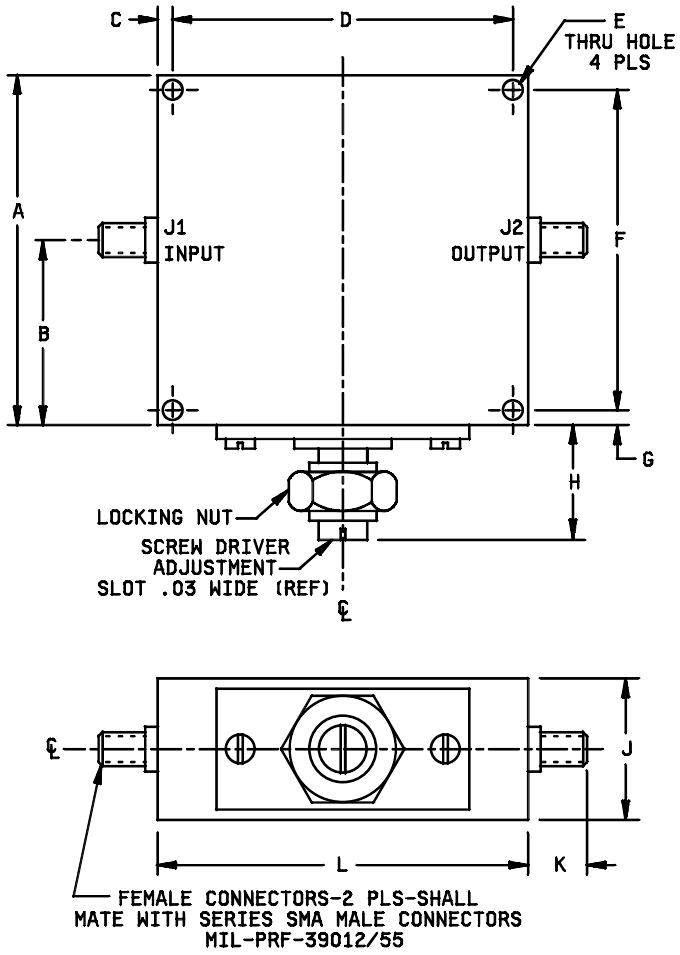
| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | 2.260 | 2.240 | 57.4 | 56.9 |
| B | .27 | .23 | 6.9 | 5.8 |
| C | 2.02 dia | 1.98 dia | 51.3 dia | 50.3 dia |
| D | .27 | .23 | 6.9 | 5.8 |
| E | 1.197 | 1.177 | 30.40 | 29.90 |
| F | .49 | .45 | 12.4 | 11.4 |
| G | .68 | .64 | 17.3 | 16.3 |
| H | 1.40 | 1.36 | 35.6 | 34.5 |
| J | 1.885 | 1.865 | 47.88 | 47.37 |
| K | 2.77 | 2.73 | 70.4 | 69.34 |
| L | .46 | .42 | 11.7 | 10.7 |
| M | .68 | .64 | 17.3 | 16.3 |
| N | .14 | .10 | 3.6 | 2.5 |
| P | 1.885 | 1.865 | 47.9 | 47.4 |
| Q | 1.03 | .97 | 26.2 | 24.6 |
| R | 1.08 | 1.04 | 27.4 | 26.4 |
| S | .90 | .86 | 22.9 | 21.8 |
| T | 2.14 | 2.10 | 54.4 | 53.3 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.02 (.5 mm).
4. Dial shall be marked in one dB steps to the 10 dB level, and in two dB steps through the 30 dB level.

FIGURE 4. Dimensions and configuration for part number M24215/9-004 - Continued.



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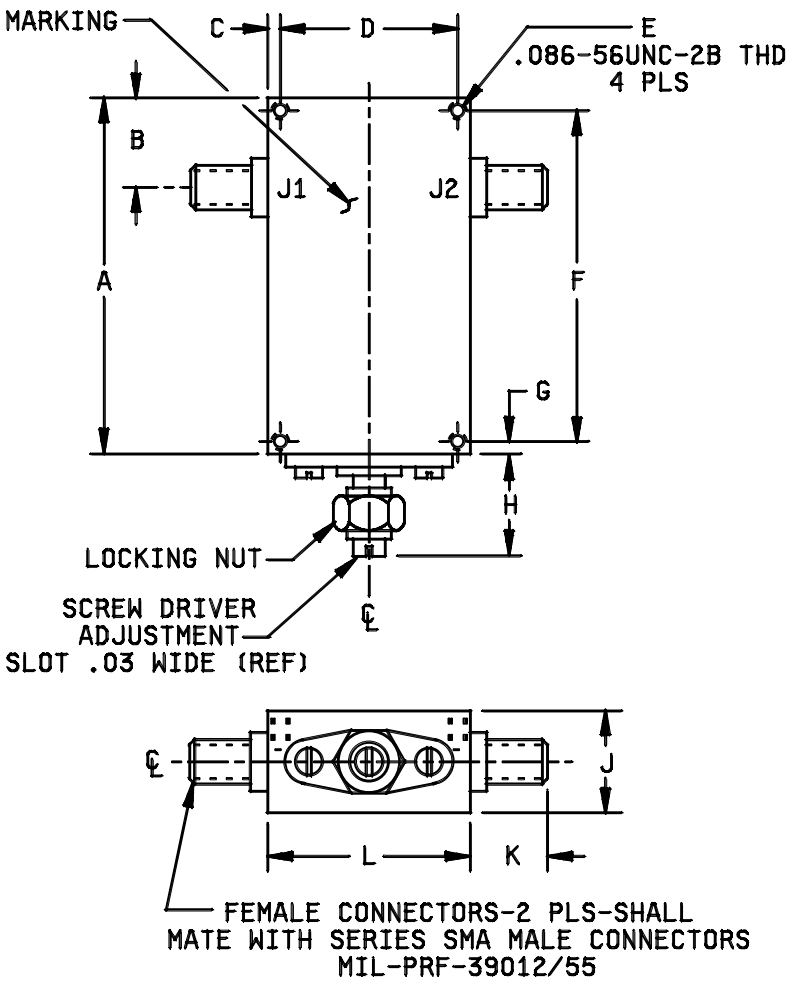

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | 2.28 | 2.22 | 57.9 | 56.4 |
| B | 1.22 | 1.16 | 31.0 | 29.5 |
| C | .12 | .06 | 3.0 | 1.5 |
| D | 2.192 | 2.182 | 55.7 | 55.4 |
| E | .133 dia | .127 dia | 3.38 dia | 3.23 dia |
| F | 2.067 | 2.057 | 52.5 | 52.2 |
| G | .12 | .07 | 3.0 | 1.8 |
| H | .65 | .59 | 16.5 | 15.0 |
| J | .94 | .88 | 23.9 | 22.4 |
| K | .41 | .35 | 10.4 | 8.9 |
| L | 2.41 | 2.35 | 61.2 | 59.7 |

FIGURE 5. Dimensions and configuration for part number M24215/9-005



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

| | Inches | | Millimeters | |
|---|---|---|---|---|
| | Max | Min | Max | Min |
| A | 1.78 | 1.72 | 45.2 | 43.7 |
| B | .47 | .41 | 11.9 | 10.4 |
| C | .09 | .03 | 2.3 | .8 |
| D | .905 | .845 | 22.99 | 21.46 |
| E | --- | .190 deep | --- | 4.83 deep |
| F | 1.655 | 1.595 | 42.04 | 40.51 |
| G | .09 | .03 | 2.3 | .8 |
| H | .53 | .47 | 13.5 | 11.9 |
| J | .53 | .47 | 13.5 | 11.9 |
| K | .41 | .35 | 10.4 | 8.9 |
| L | 1.03 | .97 | 26.2 | 24.6 |

FIGURE 6. Dimensions and configuration for part number M24215/9-006.



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REQUIREMENTS

Dimensions and configuration: See figures 1, 2, 3, 4, 5, and 6.

Nominal impedance: 50 ohms.

Final finish: Flat black lacquer, color 37038 per FED-STD-595.

Electrical requirements: See table I.

Part number: See table I.

TABLE I. Electrical requirements.

| Part number M24215/9- | Figure | Freq. range (GHz) | Atten. range (dB) 3/ | VSWR (max) | Insertion loss (max at min. atten. setting) (dB) | Calibration accuracy (dB) | Frequency sensitivity (dB) | Min input (watts) | | Nominal operating temp. range (°C) | Weight (max) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | av. | pk. | | lbs | (g) | oz |
| 001 | 1 | 2-8 | 0-15 at 2 GHz 0-20 at 4-8 GHz | 1.50:1 | 0.5 | N/A | --- | 2 | --- | -45 - +100 (See notes 1 and 2) | .16 | (73) | (2.5) |
| 002 | 2 | 2-8 | 0-15 at 2 GHz 0-20 at 4-8 GHz | 1.50:1 | 0.5 | N/A | --- | 2 | --- | -45 - +100 (See notes 1 and 2) | .16 | (73) | (2.5) |
| 003 | 3 | 7-11 | 0-20 | 1.50:1 | 0.5 | N/A | ±1.5 | 2 | --- | -45 - +100 (See notes 1 and 2) | .16 | (73) | (2.5) |
| 004 | 4 | 9.0-9.2 | 0-30 | 1.50:1 | 0.7 | ±0.5 | ±0.3 | 2 | 200 | -18 - +70 | 1 | (454) | (16) |
| 005 | 5 | 4.0-18 | 0-15 | 1.50:1 | 1.0 | N/A | ±2 | 5 | 3000 | -30 - +105 | .5 | (227) | (8) |
| 006 | 6 | 11-17 | 0-20 | 1.50:1 | 0.5 | N/A | ±2 | 5 | 3000 | -30 - +105 | .16 | (73) | (2.5) |

NOTES:

1. Non-operating ambient temperature range: -60°C to +110°C.
2. Temperature coefficient: 1.0 dB maximum change over the temperature range -45°C to +55°C at any attenuation setting.
3. Continuously variable.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS
Air Force - 99

Preparing activity:
DLA - CC

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